

The Hall Matching Problem

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1 Problem Statement

Problem. *We are given a set of mice and a set of cheeses, and each mouse likes some of the cheeses. Can we give each mouse a cheese it likes to eat?*

Let us introduce notation to make this easier. We will denote the set of all mice by $\mathcal{M}$ and the set of all cheeses by $\mathcal{C}$. We will have a function $L : \mathcal{M} \rightarrow \mathcal{P}(\mathcal{C})$ that takes a mouse m to the subset of cheeses that it likes. That is, given a mouse m , $L(m)$ is the set of cheeses that it likes.

We want to find an eating function $E : \mathcal{M} \rightarrow \mathcal{C}$. This should be injective, as we cannot give the same cheese to two mice. We should also have that $E(m) \in L(m)$ for all mice m (in other words, each mouse should like the cheese that it eats). This gives us a more precise way to pose our problem.

Problem. *Given sets $\mathcal{M}$ and $\mathcal{C}$ along with a function $L : \mathcal{M} \rightarrow \mathcal{P}(\mathcal{C})$, does there exist an injective function $E : \mathcal{M} \rightarrow \mathcal{C}$ such that $E(m) \in L(m)$ for all $m \in \mathcal{M}$?*

We can find the solution to this problem.

Theorem. *Given sets $\mathcal{M}$ and $\mathcal{C}$ along with a function $L : \mathcal{M} \rightarrow \mathcal{P}(\mathcal{C})$, there exists an injective function $E : \mathcal{M} \rightarrow \mathcal{C}$ such that $E(m) \in L(m)$ for all $m \in \mathcal{M}$ if and only if, for all subsets $M \subseteq \mathcal{M}$, $|M| \leq |\bigcup_{m \in M} L(m)|$.*

The only if direction is fairly clear. If there exists such a function E , then $E(M) \subseteq \bigcup_{m \in M} L(m)$, so $|M| = |E(M)| \leq |\bigcup_{m \in M} L(m)|$. We only have to prove the if direction. We introduce some notation to make this easier.

Notation. Given a subset $M \subseteq \mathcal{M}$, we write $L(M) = \bigcup_{m \in M} L(m)$. If a mouse m likes a cheese c , we write $m \rightarrow c$ or $c \rightarrow m$.

Note that this use of $L(M)$ is slightly different from the convention of $\bigcup_{m \in M} \{L(m)\}$. Rather than being a subset of $\mathcal{P}(\mathcal{C})$, $L(M)$ is an element of $\mathcal{P}(\mathcal{C})$. We will never use the original convention here, so there will be no ambiguity.

2 Non-Constructive Proof

We first present a non-constructive proof.

Proof. We will use strong induction on $|\mathcal{M}|$. The base case is $|\mathcal{M}| = 0$, so that $\mathcal{M} = \emptyset$. The condition holds, as $\emptyset$ is the only subset of $\emptyset$, and $|\emptyset| = 0 \leq 0 = |L(\emptyset)|$. We can vacuously find an injection E , so the base case holds. Now suppose that $|\mathcal{M}| = n$, and that we can find an injection E whenever $|\mathcal{M}| < n$ and the condition holds. We have two cases.

First, suppose that there exists an $\emptyset \neq M \subsetneq \mathcal{M}$ with $|M| = |L(M)|$. Since $M \subseteq \mathcal{M}$, the condition also holds on

$$\begin{aligned} L_M : M &\rightarrow \mathcal{P}(L(M)) \subseteq \mathcal{P}(\mathcal{C}) \\ m &\mapsto L(m). \end{aligned}$$

We claim that the condition will also hold on

$$\begin{aligned} L_{\mathcal{M} \setminus M} : \mathcal{M} \setminus M &\rightarrow \mathcal{P}(\mathcal{C} \setminus L(M)) \\ m &\mapsto L(m) \setminus L(M). \end{aligned}$$

Take $X \subseteq \mathcal{M} \setminus M$, and consider $X \cup M \subseteq \mathcal{M}$. We have that

$$\begin{aligned} |X| + |M| &= |X \cup M| \leq \left| \bigcup_{m \in X \cup M} L(m) \right| = \left| \bigcup_{m \in X} L(m) \cup \bigcup_{m \in M} L(m) \right| = \left| \bigcup_{m \in X} L(m) \setminus L(M) \cup \bigcup_{m \in M} L(m) \right| \\ &= \left| \bigcup_{m \in X} L_{\mathcal{M} \setminus M}(m) \cup \bigcup_{m \in M} L(m) \right| = \left| \bigcup_{m \in X} L_{\mathcal{M} \setminus M}(m) \right| + \left| \bigcup_{m \in M} L(m) \right| \\ &= \left| \bigcup_{m \in X} L_{\mathcal{M} \setminus M}(m) \right| + |L(M)| = \left| \bigcup_{m \in X} L_{\mathcal{M} \setminus M}(m) \right| + |M|. \end{aligned}$$

Thus,

$$|X| \leq \left| \bigcup_{m \in X} L_{\mathcal{M} \setminus M}(m) \right|,$$

so the condition holds.

Now, by the inductive hypothesis, we can find two injections $E_M : M \rightarrow L(M)$ and $E_{\mathcal{M} \setminus M} : \mathcal{M} \setminus M \rightarrow \mathcal{C} \setminus L(M)$ such that $E_M(m) \in L_M(m) = L(m)$ and $E_{\mathcal{M} \setminus M}(m) \in L_{\mathcal{M} \setminus M}(m) \subseteq L(m)$. Thus, if we define $E : \mathcal{M} \rightarrow \mathcal{C}$ by

$$E(m) = \begin{cases} E_M(m) & m \in M \\ E_{\mathcal{M} \setminus M}(m) & m \notin M \end{cases},$$

this is an injection satisfying $E(m) \in L(m)$.

Suppose now that no such M exists. Since the condition holds on $\mathcal{M}$, we have that $|M| < |L(M)|$ for all $\emptyset \neq M \subsetneq \mathcal{M}$. Fix $m_0 \in \mathcal{M}$ and $c \in L(m_0)$. Consider the map

$$\begin{aligned} L_{\mathcal{M} \setminus \{m_0\}} : \mathcal{M} \setminus \{m_0\} &\rightarrow \mathcal{P}(\mathcal{C} \setminus \{c\}) \\ m &\mapsto L(m) \setminus \{c\}. \end{aligned}$$

We claim that the condition holds on this. Indeed, for any $\emptyset \neq M \subseteq \mathcal{M} \setminus \{m_0\}$,

$$|M| < |L(M)| \leq |L_{\mathcal{M} \setminus \{m_0\}}(M) \cup \{c\}| = |L_{\mathcal{M} \setminus \{m_0\}}(M)| + 1.$$

Since we have a strict inequality and are working with integers, we can subtract 1 to get $|M| \leq |L_{\mathcal{M} \setminus \{m_0\}}(M)|$. If $M = \emptyset$, we have that $|M| = 0 \leq 0 = |L_{\mathcal{M} \setminus \{m_0\}}(M)|$. Hence, the condition holds, and we can find an

injection $E_{\mathcal{M} \setminus \{m_0\}} : \mathcal{M} \setminus \{m_0\} \rightarrow \mathcal{C} \setminus \{c\}$ with $E_{\mathcal{M} \setminus \{m_0\}}(m) \in L_{\mathcal{M} \setminus \{m_0\}}(m) \subseteq L(m)$ by the inductive hypothesis. If we define

$$E(m) = \begin{cases} E_{\mathcal{M} \setminus \{m_0\}}(m) & m \neq m_0 \\ c & m = m_0 \end{cases},$$

we have found an injection from $\mathcal{M}$ to $\mathcal{C}$ with $E(m) \in L(m)$ for all m . ■

3 Constructive Proof

We now present a constructive proof. We follow the following algorithm:

Algorithm (Main Algorithm). Start the below algorithm with $\mathcal{M} = \{A\}$, where E is the empty injection.

- (1) Start with a total set of mice $\mathcal{M}$ containing a mouse to add A , where we have an injection $E : \mathcal{M} \setminus \{A\} \rightarrow \mathcal{C}$. Take $C = \text{im } E$.
- (2) Construct sets

$$\{A\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_{\ell-1} \subsetneq M_\ell \subseteq \mathcal{M}$$

using Sub-Algorithm 1.

- (3) Take $N_i = M_i \setminus M_{i-1}$ for $0 < i \leq \ell$ and $N_0 = \{A\}$.
- (4) Take an element $m_\ell \in M_\ell$ such that there is $c \in \mathcal{C} \setminus C$ with $m_\ell \rightarrow c$.
- (5) Find a chain

$$A = m_0 \rightarrow E(m_1) \leftarrow m_1 \rightarrow E(m_2) \leftarrow m_2 \rightarrow \cdots \leftarrow m_{\ell-1} \rightarrow E(m_\ell) \leftarrow m_\ell$$

using Sub-Algorithm 2.

- (6) Define $E' : \mathcal{M} \rightarrow \mathcal{C}$ by

$$E'(m) = \begin{cases} c & m = m_\ell \\ E(m_{i+1}) & m = m_i \ (0 \leq i < \ell) \\ E(m) & \text{else} \end{cases}$$

- (7) If there are no more mice to add, output E' .
- (8) Take a mouse to add A' . Go to (1) with $\mathcal{M}' = \mathcal{M} \cup \{A'\}$, A' and E' .

Sub-Algorithm 1. Start the below algorithm with $M_0 = \{A\}$ and $\ell = 0$.

- (1) Start with

$$\{A\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_\ell \subseteq \mathcal{M}.$$

- (2) If $L(M_\ell) \not\subseteq C$, output these sets.
- (3) Go to (1) with sets

$$\{A\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_\ell \subsetneq M_{\ell+1} = E^{-1}(L(M_\ell)) \cup \{A\} \subseteq \mathcal{M}.$$

Sub-Algorithm 2. Start the below algorithm with $m_\ell \in N_\ell$.

- (1) Start with $m_i \in N_i$
- (2) If $i = 0$, output m_i .
- (3) Take $m_{i-1} \in M_{i-1}$ with $E(m_i) \in L(m_{i-1})$.
- (4) Go to (1) with m_{i-1} to get a chain

$$A = m_0 \rightarrow E(m_1) \leftarrow m_1 \rightarrow E(m_2) \leftarrow m_2 \rightarrow \cdots \leftarrow m_{i-2} \rightarrow E(m_{i-1}) \leftarrow m_{i-1}.$$

- (5) Extend the chain to

$$A = m_0 \rightarrow E(m_1) \leftarrow m_1 \rightarrow E(m_2) \leftarrow m_2 \rightarrow \cdots \leftarrow m_{i-1} \rightarrow E(m_i) \leftarrow m_i.$$

We must show that this algorithm halts, is executable, and provides a valid injection. We use the following lemmas.

Lemma. *Step (3) of Sub-Algorithm 1 provides a valid input to Sub-Algorithm 1.*

Proof. If $m \in M_i \setminus \{A\}$, then $E(m) \leftarrow m$ so that $E(m) \in L(M_i)$. Thus, $m \in E^{-1}(L(M_i)) \subseteq M_{i+1}$. Since $A \in M_{i+1}$, we have that $M_i \subseteq M_{i+1}$. Now, note that $|M_{i+1}| = 1 + |E^{-1}(L(M_i))| = 1 + |L(M_i)|$. By the condition, $1 + |L(M_i)| \geq 1 + |M_i| > |M_i|$, so we have that $M_{i+1} \neq M_i$. Thus, $M_i \subsetneq M_{i+1}$. ■

Lemma. *Sub-Algorithm 1 halts.*

Proof. Notice that there must be some mouse $m \in \mathcal{M}$ that likes some cheese $c \notin C$. If not, all mice would only like cheeses in C . However, $C = E(\mathcal{M} \setminus \{A\})$, so $|C| = |\mathcal{M}| - 1$, which violates the condition. Now, the Sub-Algorithm 1 constructs larger and larger subsets of $\mathcal{M}$. Since $\mathcal{M}$ is finite, and we know that at least one such mouse exists, the subsets must eventually contain such a mouse. Hence, the process will halt. ■

Lemma. *The output of Sub-Algorithm 1 gives sets such that $L(M_i) \subseteq C$ and $L(M_i) = E(M_{i+1} \setminus \{A\})$ for $1 \leq i < \ell$, and there is some $m \in M_\ell$ such that $L(m) \cap (C \setminus C) \neq \emptyset$.*

Proof. These can all be seen with the construction provided by Sub-Algorithm 1 ■

Lemma. *The value m_ℓ in Main Algorithm (4) is in N_ℓ .*

Proof. $L(M_{\ell-1}) \subseteq C$, but $L(m_\ell) \not\subseteq C$. Thus, $m_\ell \notin M_{\ell-1}$. ■

Lemma. *The value m_{i-1} in Sub-Algorithm 2 (3) is an element of N_{i-1} .*

Proof. Suppose instead that $m_{i-1} \notin N_{i-1}$, so that $m_{i-1} \in M_{i-2}$. Then, $E(m_i) \in L(m_{i-1}) \subseteq L(M_{i-2})$, meaning that $m_i \in E^{-1}(L(M_{i-2})) = M_{i-1} \setminus \{A\}$, contradicting the fact that $m_i \in N_i$. Thus, $m_{i-1} \in N_{i-1}$. ■

Lemma. *Main Algorithm (6) gives an injective function E' .*

Proof. Notice first that $E(m_1), \dots, E(m_\ell), c$ are all distinct. Call this set C' . Notice also that we never map to an element of C' in the “else” case, as the only way to map to these through E are through elements $m_1, \dots, m_\ell$ that are considered before this case.

Now suppose that $E'(m_1) = E'(m_2)$. First suppose that $E'(m_1) = E'(m_2) \in C'$. If they are both c , $m_1 = m_2 = m_\ell$. If they are both $E(m_{i+1})$ for some $0 \leq i < \ell$, we must have $m_1 = m_2 = m_i$. Thus, E' is injective in this case.

Now suppose that $E'(m_1) = E'(m_2) \notin C'$. Then $E(m_1) = E'(m_1) = E'(m_2) = E(m_2)$, so $m_1 = m_2$ since E is injective. ■

4 Graph-Theoretic Proof

We reformulate the problem as follows: suppose that G is a finite bipartite graph on $\mathcal{M}$ and $\mathcal{C}$. We want to find a matching that covers every vertex in $\mathcal{M}$ (call this $\mathcal{M}$ -perfect). If, for $M \subseteq \mathcal{M}$, $L(M)$ denotes the neighborhood of M , then the condition is $|M| \leq |L(M)|$ for all subsets $M \subseteq \mathcal{M}$.

Proof. Suppose that there is no $\mathcal{M}$ -perfect matching. Let E be a matching that matches a maximum number of elements in $\mathcal{M}$. Let A be a vertex in $\mathcal{M}$ that is not matched by E . Consider the set of all paths in G starting from A that alternate between using edges in E and edges not in E (these paths all start with an edge not in E , in particular). If we consider the direction of these edges, all edges from $\mathcal{M}$ to $\mathcal{C}$ are not in E , and all edges from $\mathcal{C}$ to $\mathcal{M}$ are in E . Let the set of all vertices in $\mathcal{C}$ connected to A by one of these paths be C , and the set of all vertices in $\mathcal{M}$ connected to A by one of these paths (including A itself) be M . Suppose that we have a maximal alternating path ending at some element in $\mathcal{C}$. Since all edges in this path towards $\mathcal{C}$ are not in E , and all edges away from $\mathcal{C}$ are in E , we can switch whether the edges in this path are in E to get a new matching that matches A along with all the elements matched by E , contradicting E 's maximality. Hence, all maximal alternating paths end at a vertex in $\mathcal{M}$, and so in particular, every vertex in C is matched by E to some vertex in $M \setminus \{A\}$ (namely, the vertex after it in the maximal alternating path passing through it). On the other hand, every vertex in $M \setminus \{A\}$ is matched by E to some vertex in C (the vertex before it in an alternating path passing through it). Thus, E gives a bijection between $M \setminus \{A\}$ and C , so $|C| = |M| - 1$. Now, if $v \in L(M)$ is connected to $m \in M$ by an edge in E , it comes before m in any alternating path from A to m , so that $v \in C$. If the edge between v and m is not in E , we can extend an alternating path from A to m to get an alternating path from A to v . Thus, $v \in C$ so that $L(M) \subseteq C$. We therefore have that

$$|L(M)| \leq |C| = |M| - 1 < |M|,$$

so the condition does not hold. ■